

Maximum capillary pressure of particle-laden emulsion films: monolayers, bilayers and defects

Pore-scale model of film rupture by capillary penetration through the gaps of a particle layer. Every curve comes from one closed-form or numerical model with no fitted parameters (code: *pore_model.py*, *fig_pmax_integrated.py*). October 2026.

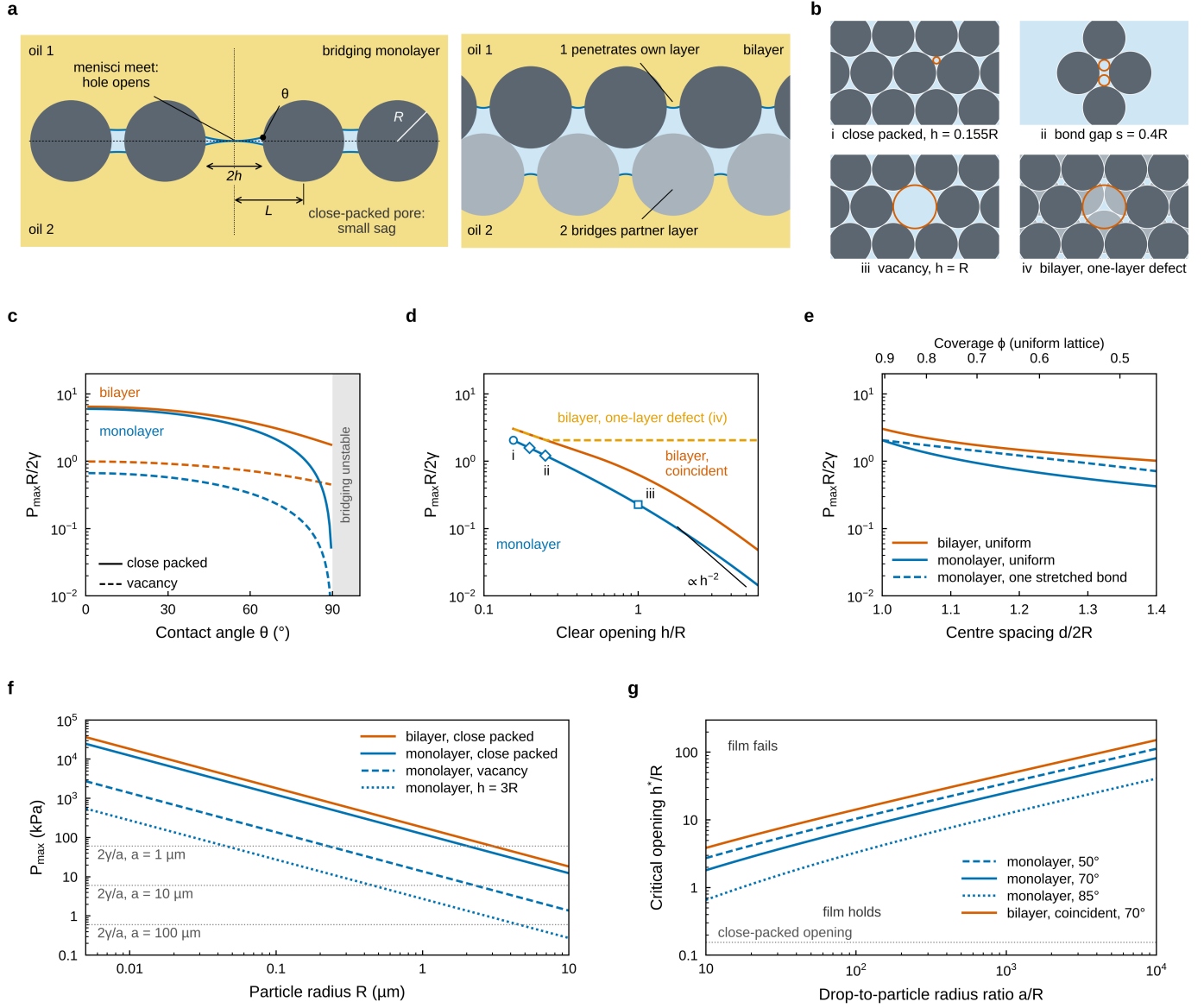


Fig. 1 | The largest gap in the particle layer, not the close-packed lattice, sets the maximum capillary pressure of a Pickering emulsion film. **a**, Cross-sections at the onset of failure, drawn to scale from the model ($\theta = 70^\circ$). Left: a bridging monolayer whose central pore is widened to a clear opening $h = 0.7R$ between close-packed neighbours. At $P = P_{\max}$ the two oil-water menisci (blue) touch on the mid-plane (dotted) and a hole opens, while the menisci in the close-packed pores have barely sagged. L , distance from the pore axis to the particle centres; θ , contact angle measured through water. Right: a nested bilayer loaded to 70% of its failure pressure. Oil 1 must first penetrate its own layer (1); the partner layer then behaves as a bridging monolayer (2). **b**, Top views of the pore geometries used below; the orange circle is the largest particle-free disc (radius h) in each pore. In iv the vacancy in the upper layer (dark) is covered by the intact lower layer (light). **c**, Dimensionless failure pressure $P_{\max}R/2\gamma$ against contact angle for a close-packed lattice (solid) and a single vacancy (dashed). The monolayer value vanishes as $\cos \theta$; the bilayer does not. For $\theta > 90^\circ$ (grey) a bridging particle dewets and ruptures the film. **d**, Failure pressure against clear opening h/R ($\theta = 70^\circ$). Symbols on the monolayer curve mark the geometries of b (i, close packed; ii, bond gaps $s = 0.2R$ and $0.4R$; iii, vacancy). The monolayer follows the closed form $4\gamma R \cos \theta / [h(h + 2R)]$, which tends to h^{-2} for wide gaps. A defect confined to one layer of a bilayer (orange dashed) cannot lower $P_{\max}$ below the bridging value of the intact partner layer. **e**, Failure pressure against centre-to-centre spacing d for a uniformly expanded hexagonal lattice (top axis: area coverage $\phi = 0.907(2R/d)^2$) and for one stretched bond in an otherwise close-packed layer. The axis range matches c and d. **f**, Absolute $P_{\max}$ against particle radius for $\gamma = 30 \text{ mN m}^{-1}$ and $\theta = 70^\circ$ (all curves scale as γ/R). Dotted lines: Laplace pressure $2\gamma/a$ of drops of radius a , the scale of the film capillary pressure in a quiescent cream. **g**, Critical clear opening h^* at which $P_{\max}$

falls to $2\gamma/a$, against the drop-to-particle size ratio. γ cancels, so the curves hold for any oil. Films whose largest opening is smaller than h^* hold. In c–g the bilayer is a nested (hcp) stack with layer spacing $1.63R$. No experimental data are plotted; the reported critical pressures we found are discussed in SI Note 3.

Focus and logic arc

Focus. A particle layer holds two oils apart because each oil–water meniscus must bulge through the gaps between particles before the two oils can touch. The figure makes one claim. The pressure a film can withstand is set by its widest gap, through a single closed-form law, $P_{\max} = 4\gamma R \cos \theta / [h(h + 2R)]$. A second particle layer changes the answer qualitatively, because a gap in one layer is covered by the other.

a → b: mechanism and geometry. Panel a shows the failure event itself rather than an idealized cartoon. Both sketches are drawn from the model, so the sag of each meniscus is to scale. A widened pore fails while its close-packed neighbours are still nearly flat, which is why the widest gap matters. Panel b defines the four geometries used in the plots, each by a single number: the radius h of the largest particle-free disc.

c: contact angle separates monolayer from bilayer. For a bridging monolayer the menisci only need to reach the mid-plane, and the required curvature goes to zero as $\cos \theta$. A bilayer meniscus must instead push past the equator of its own particles before it can reach the other oil. That threshold stays finite at $\theta = 90^\circ$. At $\theta = 70^\circ$ the close-packed bilayer is $1.48\times$ stronger than the monolayer; at 85° it is $3.8\times$ stronger. This is the contact-angle signature that distinguishes the two structures in an experiment.

d: the central panel. One curve covers every defect: $P_{\max} \propto 1/h$ for narrow gaps and $\propto 1/h^2$ for wide ones. A single vacancy lowers $P_{\max}$ of a monolayer 9-fold. Stretching one bond by $0.2R$ already costs 23%, independent of θ and γ . In a bilayer, a vacancy in one layer leaves $P_{\max}$ at the intact partner's bridging value (2.05 versus 3.04 in units of $2\gamma/R$, about -32%). Only coincident defects reach the low values (0.63). Bilayers are therefore defect tolerant and monolayers are not.

e: spacing. This panel separates two situations that are often lumped together as 'lower coverage'. Expanding the whole lattice (for example by electrostatic repulsion between particles) and stretching one bond lower $P_{\max}$ by similar amounts at small strain. Uniform expansion keeps lowering it, because every pore widens at once. Coverage alone therefore does not set $P_{\max}$; the size of the largest opening does.

f: absolute scale. Converted to kilopascals for $\gamma = 30 \text{ mN m}^{-1}$, a defect-free layer of $0.5 \mu\text{m}$ particles holds 246 kPa (monolayer) or 365 kPa (bilayer). That is far above the 6 kPa Laplace pressure of a $10 \mu\text{m}$ drop. A perfect lattice is therefore almost never the limiting element. A vacancy (27 kPa) or a gap three particle radii wide (5.5 kPa) brings $P_{\max}$ into the range of real loads.

g: the design rule. Equating $P_{\max}$ with the drop Laplace pressure gives a critical opening, $h^*(h^* + 2R) = 2aR \cos \theta$. γ drops out, so this holds for any oil. For $a/R = 20$ ($0.5 \mu\text{m}$ particles, $10 \mu\text{m}$ drops) and $\theta = 70^\circ$, a monolayer tolerates bare patches up to $h^* \approx 2.8R$, about three particle diameters across. A bilayer requires two such patches aligned. In practice, coalescence of particle-coated drops is a question of coverage defects, not of the close-packed lattice.

What was left out, and why

Measured critical pressures (silica/decane $\approx 100 \text{ kPa}$; carbon black 2.2–4.6 kPa) are not plotted. They are centrifuge onset pressures (osmotic pressure at the base of a cream), not film capillary pressures. No particle radius was reported, and we could only find them in secondary sources. Placing them on panel f would require an invented x-coordinate (SI Note 3). **The empirical prefactor band** $p = 0.1\text{--}1$ in Kaptay's form $2p\gamma \cos \theta/R$ is not drawn. We could not check it against Kaptay's tabulated values. The model instead gives $p_{\text{eff}} = 2R^2/[h(h + 2R)]$ (SI Note 1), so $p = 0.1\text{--}1$ corresponds to openings of $0.7R\text{--}3.6R$, which is consistent with the defect picture. **Particle detachment** is excluded: in a bridging monolayer both oils press on a particle with the same pressure, so there is no net normal force. **Water-in-oil emulsions** follow from $\theta \rightarrow 180^\circ - \theta$ and add nothing new. **Drainage kinetics, disjoining pressure and contact-angle hysteresis** change when a film fails, not the static threshold. They are discussed in SI Note 2.

Supplementary Information

SI Note 1 | Pore model and equations

Consider the opening between particles in an oil-water film. Each particle is a sphere of radius R . The pore is replaced by an axisymmetric ring of spheres whose centres lie a distance L from the pore axis, so the largest particle-free disc in the centre plane has radius $h = L - R$. For the triangular pore of a close-packed layer, $L = 2R/\sqrt{3}$ and $h = 0.155R$ exactly. The oil pressure exceeds the water pressure by P , so the meniscus in the pore is a spherical cap of radius $R_c = 2\gamma/P$ that bulges into the water. It meets each particle at the contact angle θ (measured through water). If the cap slope at the contact line is ψ , the contact line sits at polar angle $\alpha = \theta + \psi$ on the sphere, and the cap spans a circle of radius $\rho = L - R \sin(\theta + \psi)$. Laplace's law then gives

$$PR/2\gamma = \sin \psi / [L/R - \sin(\theta + \psi)].$$

As P rises, ψ grows and the cap sags by $\rho \tan(\psi/2)$ below its contact line. The film fails at the first of two events. *Meet*: the sag reaches the plane where the opposing meniscus lies. *Slide*: P passes the maximum of the expression above, so the contact line can no longer stay put and slides down the particle. For a bridging monolayer the two menisci meet on the centre plane. At that moment the cap sphere touches the centre plane on the pore axis, so its centre is at height R_c . It also meets each particle sphere at angle θ . The law of cosines for the two intersecting spheres gives $L^2 + R_c^2 = R^2 + R_c^2 + 2RR_c \cos \theta$, and hence the closed form

$$P_{\max} = 4\gamma R \cos \theta / (L^2 - R^2) = 4\gamma R \cos \theta / [h(h + 2R)] \text{ (bridging monolayer).}$$

A full numerical solution of the cap equations confirms that the menisci meet before the sliding instability for every θ from 5° to 85° and every h from $0.155R$ to $5R$. The closed form matches it to within 3×10^{-4} . Special cases follow directly. A close-packed hexagonal monolayer gives $P_{\max} = 12\gamma \cos \theta/R$. A uniformly expanded hexagonal lattice with centre spacing d gives $12\gamma R \cos \theta/(d^2 - 3R^2)$. One bond stretched to $d = 2R + s$ turns the neighbouring pore into a $(2R, 2R, d)$ triangle with circumradius $L = 4R^2/(16R^2 - d^2)^{1/2}$. This gives $P_{\max}(d)/P_{\max}(2R) = (16R^2 - d^2)/(3d^2)$, valid while $d < 2\sqrt{2}R$, independent of θ and γ . A vacancy ($h = R$) gives $4\gamma \cos \theta/3R$, one ninth of the close-packed value. A $\theta < 90^\circ$ is required: for $\theta > 90^\circ$ the expression changes sign, which is the bridging-dewetting instability of particle antifoams.

Written in Kaptay's form, $P_{\max} = 2p\gamma \cos \theta/R$, the model gives an effective prefactor $p_{\text{eff}} = 2R^2/[h(h + 2R)]$. That is 6 for a close-packed layer, 0.67 for a vacancy and 0.13 for $h = 3R$. This model is the more optimistic member of the family. Replacing three discrete spheres by a continuous ring makes the pore more confining than it really is, so absolute values for close packing are probably too high by tens of per cent. Ratios between geometries are less sensitive to this. An earlier version of this figure used an assumed band $p = 0.1$ -1. In the present picture that band corresponds to films limited by openings of $0.7R$ - $3.6R$, not by the ideal lattice.

Bilayers. Each oil is covered by its own layer, and the layers are stacked in the nested position with centre planes $1.63R$ apart. Oil 1 can only reach oil 2 in two steps. First, its meniscus must penetrate its own layer, which needs pressure P_{pen} . For close packing this is set by the sliding instability, because the meniscus cannot reach the mid-plane while still pinned above the particle equator. Second, once oil 1 has flooded the gap, the partner layer is wetted by oil on both sides. It then fails like a bridging monolayer, at P_{br} . Either oil may go first, so

$$P_{\max}^{\text{bilayer}} = \min\{ \max[P_{\text{pen}}(1), P_{\text{br}}(2)], \max[P_{\text{pen}}(2), P_{\text{br}}(1)] \}.$$

Because $P_{\text{br}} < P_{\text{pen}}$ for the same lattice, an intact bilayer fails at P_{pen} . That is $3.04 \times 2\gamma/R$ at $\theta = 70^\circ$, against 2.05 for the bridging monolayer. It remains finite at $\theta = 90^\circ$ ($\approx 1.8 \times 2\gamma/R$), which matches the qualitative statement that a double layer carries an extra positive term in Kaptay's formula. A defect in one layer only lowers the first step. The partner's bridging threshold then sets the failure pressure, so a vacancy in one layer costs 32% at $\theta = 70^\circ$. Two coincident vacancies cost 79%.

Table S1 | $P_{\max}R/2\gamma$ for the main geometries.

θ ($^\circ$)	Monolayer, close packed	Bilayer, close packed	Monolayer, vacancy	Bilayer, coincident vacancies	Bilayer, vacancy in one layer
30	5.20	5.72	0.58	0.92	5.20
50	3.86	4.52	0.43	0.80	3.86
60	3.00	3.79	0.33	0.72	3.00
70	2.05	3.04	0.23	0.63	2.05
80	1.04	2.33	0.12	0.54	1.04
85	0.52	2.01	0.06	0.49	0.52

Table S2 | Absolute $P_{\max}$ (kPa) for $\gamma = 30 \text{ mN m}^{-1}$, $\theta = 70^\circ$. For comparison, the drop Laplace pressure $2\gamma/a$ is 60, 6 and 0.6 kPa for $a = 1, 10$ and $100 \mu\text{m}$.

Particle radius	Monolayer, close packed	Bilayer, close packed	Monolayer, vacancy	Monolayer, $h = 3R$
10 nm	12,313	18,235	1,368	274
100 nm	1,231	1,824	137	27.4
0.5 μm	246	365	27.4	5.47
1 μm	123	182	13.7	2.74
5 μm	24.6	36.5	2.74	0.547

SI Note 2 | Stability of Pickering emulsions

Particles stabilize emulsions differently from surfactants. A particle of radius R at an oil-water interface lowers the free energy by $\pi R^2 \gamma (1 - |\cos \theta|)^2$. For a $0.5 \mu\text{m}$ particle at $\theta = 70^\circ$ and $\gamma = 30 \text{ mN m}^{-1}$ this is about 10^{-14} J , or $10^6 kT$. Even for 10 nm particles it is of order 10^2 – $10^3 kT$. Adsorption is therefore effectively irreversible. The layer does not relax by exchange with the bulk, as a surfactant monolayer does. Instead it acts as a solid, jammed skin that is fixed once the drop is made. Two consequences follow. Drop size is fixed by the available particle area through limited coalescence: drops merge during and after emulsification until their total area is just covered. Dense, nearly close-packed layers are therefore the natural end state, not a special case. And because the particles cannot leave, Ostwald ripening and deformation are resisted mechanically, which is why particle-coated drops can stay non-spherical (arrested coalescence).

When two coated drops are pressed together, by creaming, by centrifugation or in a concentrated emulsion, the water between them forms a film. The film can take two structures. In a bilayer each drop keeps its own layer. In a bridging monolayer a single layer of particles is shared by both drops, each particle protruding into both oils. Squeezing does not turn a close-packed bilayer into a monolayer. A monolayer holds half as many particles per area, so half of them would have to leave the interfaces, at a cost of order $10^6 kT$ each. Bridging monolayers form instead when the layers are sparse enough to interdigitate (coverage below about half of close packing), typically during emulsification or with too little particle. They are only stable for $\theta < 90^\circ$ (measured through the continuous phase). For $\theta > 90^\circ$ a bridging particle pulls the two menisci together and ruptures the film, which is the bridging-dewetting mechanism of particle antifoams. Both structures have been observed in the same emulsion.

A coated film fails when the oil reaches across it. The particles never have to move: the oil goes through the gaps between them once the capillary pressure exceeds what the curved menisci in those gaps can hold (SI Note 1). Three features matter in practice. First, the threshold scales as γ/R , so smaller particles give stronger films if they can be packed equally well. Second, it falls with contact angle as $\cos \theta$ for a bridging monolayer. Particles too close to 90° give weak bridged films, although they adsorb most strongly, while very hydrophilic particles ($\theta \lesssim 30^\circ$) adsorb weakly. This trade-off underlies the often-quoted optimum near $\theta \approx 70^\circ$ for oil-in-water emulsions. Third, and most important, the threshold is set by the widest opening, not by the average packing. The film pressure in a quiescent cream is of order the drop Laplace pressure $2\gamma/a$. That is one to three decades below the threshold of an ideal lattice, so the drops that do merge are those with uncovered patches several particle diameters wide. Such patches come from incomplete coverage, polydisperse or aggregated particles, grain boundaries in the jammed layer, and particles pushed aside by lateral capillary forces as the menisci deform.

This picture explains several observations without extra assumptions. Emulsions near the limited-coalescence boundary, where coverage is just sufficient, are fragile, while a modest excess of particle makes them very robust. Raising the particle concentration beyond full coverage improves stability only insofar as it heals defects or creates bilayers and particle networks in the continuous phase. Salt and pH, which control particle aggregation and θ , change stability by orders of magnitude, through both the contact angle and the size of the gaps in the layer. Bilayers, which form naturally between densely covered drops, are far more forgiving than shared monolayers, because a gap in one layer is covered by the other.

The model is static and deliberately minimal. It treats the pore as axisymmetric and ignores gravity and line tension. It assumes the menisci merge as soon as they touch. With surfactant or strong electrostatic repulsion, a thin water film could survive contact and add a disjoining-pressure barrier, so the meet criterion is a lower bound in that case. Contact-angle hysteresis matters because the oil advances, so the water-receding angle applies. It is smaller than the equilibrium angle and raises $P_{\max}$. Particle rearrangement under load and drainage kinetics decide *when* a film reaches its threshold, not the threshold itself. Two experiments would test the main predictions most directly. A thin-film pressure balance (porous-plate cell) applies a known capillary

pressure to a single coated film. Varying θ there should distinguish the $\cos \theta$ dependence of a bridging monolayer from the finite intercept of a bilayer. Deliberately controlled coverage should show the $1/h^2$ law of panel d.

SI Note 3 | Reported critical pressures and why they are not plotted

System	Reported value	What was measured	Status of the number
Hydrophobized silica, n-decane/water (Kruglyakov, Nushtaeva and co-workers)	≈ 100 kPa, about 25% of the authors' own calculated maximum (≈ 400 kPa)	Centrifuge onset of coalescence; also thin emulsion layers on a porous plate	Found in secondary literature; the original value was not seen. Particle radius in the test not reported to us
Carbon black 0.015 wt%, 0.6 M NaCl	4.6 kPa	Centrifuge demulsification pressure	Secondary source; original paper not confirmed
Carbon black 0.015 wt%, pH 3.3, no salt	2.2 kPa	As above	As above

A centrifuge measures the osmotic pressure at the base of the cream at the moment coalescence starts. Turning that into a film capillary pressure needs the oil fraction and drop size, which we do not have. With $0.27 \mu\text{m}$ silica particles, $\gamma = 50 \text{ mN m}^{-1}$ and $\theta \approx 70^\circ$, the model gives about 760 kPa for an ideal bridging monolayer. That is the same order as the authors' own calculated ≈ 400 kPa. The measured onset is several-fold lower, as expected if defects control failure. The carbon black values lie below every ideal-lattice prediction for any plausible particle size, and are consistent with failure at openings of a few particle radii. These are consistency checks, not tests: the values are too few and too uncertain to place on the figure.

References

1. Denkov, N. D., Ivanov, I. B., Kralchevsky, P. A. & Wasan, D. T. A possible mechanism of stabilization of emulsions by solid particles. *J. Colloid Interface Sci.* **150**, 589–593 (1992). Journal, volume and page checked; title from memory.
 2. Kaptay, G. On the equation of the maximum capillary pressure induced by solid particles to stabilize emulsions and foams and on the emulsion stability diagrams. *Colloids Surf. A* **282–283**, 387–401 (2006).
 3. Kruglyakov, P. M., Nushtaeva, A. V. & Vilkova, N. G. Experimental investigation of capillary pressure influence on breaking of emulsions stabilized by solid particles. *J. Colloid Interface Sci.* **276**, 465–474 (2004). †
 4. Horozov, T. S. & Binks, B. P. Particle-stabilized emulsions: a bilayer or a bridging monolayer? *Angew. Chem. Int. Ed.* **45**, 773–776 (2006). †
 5. Tcholakova, S., Denkov, N. D. & Lips, A. Comparison of solid particles, globular proteins and surfactants as emulsifiers. *Phys. Chem. Chem. Phys.* **10**, 1608–1627 (2008). †
 6. Pieranski, P. Two-dimensional interfacial colloidal crystals. *Phys. Rev. Lett.* **45**, 569 (1980). †
 7. Arditty, S., Whitby, C. P., Binks, B. P., Schmitt, V. & Leal-Calderon, F. Some general features of limited coalescence in solid-stabilized emulsions. *Eur. Phys. J. E* **11**, 273–281 (2003). †
 8. Binks, B. P. Particles as surfactants: similarities and differences. *Curr. Opin. Colloid Interface Sci.* **7**, 21–41 (2002). †
- † Cited from memory; details could not be checked from this environment and should be verified before use. Kaptay (ref. 2) was checked by title and volume only; its tabulated p values could not be read.